SAÉ 1.01: display of a webpage using Java

Introduction

Yesterday, for our SAE, we started a project about the development of a program in the Java language, which we shall name a "class." This class enables its users to manage the grades of students using a matrix. The technical purpose of the class is to generate a webpage (an HTML-like one) which contains a spreadsheet filled with particular data.

This data is:

- A list of subjects
- A list of students having these subjects
- Decimal grades, matching a given student's name with a subject's name.

Chronology in the execution of the program/ method

This program first lets the user input the number of subjects and students, which creates the right number of rows and columns in the spreadsheet.

It then lets the user input the labels of the spreadsheet, corresponding with the names of the subjects and of the students.

It then lets the user input the marks themselves.

It finally calculates the general average grade of every student.

The extra mandatory features of the page were to enable the user to:

- Save the whole matrix of data as a TXT file.
- Switch between the read-only and the edition modes.

As extra tasks, we chose to enable the user to:

- Save the same matrix into a CSV file.
- Rank the students according to their average grade.

Regarding the organization of the work

We have divided the whole project into several tasks which were equivalent to the amount of work to provide. Afterwards, we dispatched them among us, the three members of the group, with respect to our best capacities.

Benjamin

- I created the whole skeleton of our Java class to start on a stable basis.
- In this skeleton, I then declared a list of methods. I first let them empty (without implementing them) in order to read them as if they were a checklist to tick. Therefore, a defined method equals a method we shall implement later.
- I implemented methods which ask the user to input data. The implemented input methods are for the following: number of subjects, number of students, name of every subject, name of every student (as a whole, single entry, for example: Benjamin Boutout instead of Benjamin and Boutout).
- I also did the integration of Text Blocks containing HTML code.

Louis

Personally, I took care of the implementation of the export to CSV program, as well as the program that loads into the website a previous table that is in a TXT file.

Finally, I improved the visuals of the page through modification of the CSS program in the Text Blocks.

Marie

I implemented the Java methods enabling the management of user inputs. As a reminder, a Java method is a function or procedure within the Java class / program.

As soon as all these methods have been implemented (when our checklist lines have been ticked), I relocated the "throws IOException" segments within the code after every method declaration part. I also relocated the "nav.endPage()" with respect to the loops positions in order to avoid the connection interruptions in the web browser. To summerize, solving of the compilation and execution errors in the shell, given the detected page loading failures in the web browser.

General assessment

The SAE, in its globality, was interesting, and we all learned something despite our different skill levels. But it was a pity we didn't have enough time to correctly complete the jobs. In fact, it was stressful, especially at the end, and we had to allow some errors or inconsistencies into the code to deliver the project on time. Except for that, the team's skills have been improved, and we left this SAE with solid expertise that can be useful for our future careers. The teamwork was flowing, we all listened to each other and acted accordingly. With that work, we saw how to use a web navigator from a simplified version. Nowadays, we don't manage the navigator like that thanks to technological advances, but it's important to understand that concept. Next time, when we open our favourite navigator, we will be able to have an overview of its behaviour. Finally, in spite of the difficulties, that SAE gave our team a boost for the next part of our professional development.

Individual reports

Benjamin

To begin, I didn't find the work easy, especially with the allocated time. Indeed, the subject matter wasn't complicated, but in my opinion, we didn't handle the tools used in the course enough. Apart from that, the SAE was instructive for my personal development. My IT skills upgraded in order to create a solid base for the next part of the trip. Month after month, I can see my progress. The SAE helped me to test my expertise in real-time, like in a firm. Naturally, I'm not an expert in that area, but now I'm able to understand the general idea of that subject and work on this on my own. During my free time, I will surely explore that huge flow of knowledge for following my apprenticeship. In conclusion, the SAE was very useful for preparing me for my professional life. I don't regret having spent several hours of my life on that subject, which will be useful later.

Louis

For me, it was really instructive, first of all, about working as a team. It made me learn and remember how much you must work with the difficulties that people in the team have.

The SAE also made me learn the importance of testing what you do. For example, when we tried our program, we found out we were forced to refresh the web page to go to the next one.

On top of that, I kept on learning about programming.

Marie

In this team project, I've faced personal technical difficulties. My hard skills in software and web development are quite new and unstable, since I started to learn them less than a year ago. What's more, the first period of training in this field only lasted three months, which is very little regarding the amount of knowledge needed to be the most basic software developer. Afterwards, I landed in the IUT, where the rhythm of training became much faster, and the amount of knowledge became even greater than in my previous training. That's why, adding this parameter to my historically natural slowness, I've become slower than ever before.

On the other hand, I feel I have brought to my work group a sense of accuracy which must have been acuter thanks to my self-assessment and circumspection that I used to the benefit of my whole group.